

has the features to create a private local environment for developers to test their smart contracts before deployment.

Features

- It supports Solidity as a language for writing smart contracts.
- It provides the same features as Remix but is not accessible on the browser, at present.

Figure 4 provides compilation, running and debugging in TronStudio.

While developing dApps (distributed applications) we need to configure the nodes, peers, the database, and ordering service (order synchroniser for transactions), and then connect smart contracts with the blockchain. Setting up the network architecture manually for the application may be a hectic task, which is where frameworks are useful. We mean hectic in the sense that there are various steps involved like generating CA certificates, installing chaincodes on peer machines, and specifying the channel ID configurations which are

needed to set up and run the network. Frameworks come with default configurations and are handled using *docker-compose.yml* files, which avoids manual configuration.

Frameworks

The problem that most people face when trying to start learning something new is, where to start.

Every blockchain project has many demo applications which come with tools and libraries in a single, easily downloadable package. It already has a basic client application GUI with which users are able to interact with distributed stored smart contracts. It also has a database and Testnet configured for direct use. One can either just get familiar with the demo application, or modify it and try it for learning purposes.

Truffle boxes: This is the most popular of templates (called boilerplates) for setting up the initial structure of your dApps. Truffle boxes contain modules that take care of the

front-end (drizzle) and are connected to smart contracts, testing and deployment with an access to the blockchain framework. More details are available at <https://truffleframework.com/boxes>.

Hyperledger Umbrella: This is an open source project under the Linux Foundation and IBM. It consists of a lot of frameworks, the popular ones being Fabric, Composer and Sawtooth. These are mainly customised blockchains for enterprise use where data needs to be private, involves a lot of different members and requires transparency between enterprises.

Fabric: This has two broad aspects — infrastructure and application. It is modular in architecture and configures the required modules as and when required. The infrastructure consists of peers, endorsing peers, an ordering service and others such as MSP (membership service provider), while application components include the smart contract and the client side.

Features

- It is capable of making channels inside a network where only a member of that channel can access the related data. A member can also be a participant in two channels, depending on business needs.
- It has components like a membership service and consensus as plug-and-play components.
- It is very flexible with respect to the architecture of the network — organisations, peers belonging to organisations, etc.

Composer: This is an abstract application level framework to build and run blockchain applications on top of Hyperledger Fabric. It takes hours instead of weeks to develop an application based on the proof-of-concept.

Features

- It allows users to define the data model (.cto files), business logic and access control lists for an application that can be deployed and executed inside Fabric.

Web my-basic-sample		Define	Test	admin
PARTICIPANTS	SampleParticipant			
ASSETS	SampleAsset			
TRANSACTIONS	All Transactions			
		Date, Time	Entry Type	Participant
		2019-02-26, 10:17:56	ActivateCurrentIdentity	none
		2019-02-26, 10:17:54	StartBusinessNetwork	none
		2019-02-26, 10:17:54	IssueIdentity	none
		2019-02-26, 10:17:54	AddParticipant	none

Figure 5: Composer playground

ACCOUNTS		BLOCKS	TRANSACTIONS	LOGS
CURRENT BLOCK	0	GAS PRICE	2000000000	GAS LIMIT
			6712390	NETWORK ID
			5777	RPC SERVER
			HTTP://127.0.0.1:7545	MINING STATUS
				AUTOMINING
MNEMONIC		HD PATH		
candy naple cake sugar pudding cream honey rich smooth crumble sweet treat		m/44'/60'/0'/0'/0/account		
ADDRESS	0x627306090abaB3A6e1400e9345bC60c78a88Ef57	BALANCE	100.00 ETH	TX COUNT
				0
ADDRESS	0xf17f52151EbEF6C7334FAD080c5704D77216b732	BALANCE	100.00 ETH	TX COUNT
				0
ADDRESS	0xC5fdf4076b8F3A5357c5E395ab970B5854098Fef	BALANCE	100.00 ETH	TX COUNT
				0
ADDRESS	0x821aEa9a577a9b44299B9c15c88cf3087F3b5544	BALANCE	100.00 ETH	TX COUNT
				0

Figure 6: Gnahe Testnet shows transactions, balances and hash addresses